Available at
www.ComputerScienceWeb.com
POWERED BY SCIENCE DIRECT®

Robotics and Autonomous Systems 44 (2003) 171–179

Robotics and Autonomous Systems

www.elsevier.com/locate/robot

CPG model for autonomous decentralized multi-legged robot system—generation and transition of oscillation patterns and dynamics of oscillators

Shinkichi Inagaki [a,*], Hideo Yuasa [b,c], Tamio Arai [c,1]

[a] *Suematsu Lab., Department of Electronics-Mechanical Engineering, School of Engineering, Nagoya University, Huro-cho, Chikusa-ku, Nagoya 464-8603, Japan*

[b] *The Institute of Physical and Chemical Research (RIKEN) at Nagoya, 2271-130, Anagahora, Shimoshidami, Moriyama-ku, Nagoya 463-0003, Japan*

[c] *Arai-Yuasa-Ota Lab., Department of Precision Engineering, School of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan*

Abstract

A central pattern generator (CPG) model is proposed for the gait-pattern generation mechanism of an autonomous decentralized multi-legged robot system. The topological structure of the CPG is represented as a graph on which two time evolution systems, the Hamilton system and a gradient system, are introduced. The CPG model can generate oscillation patterns depending only on the network topology and can bifurcate different oscillation patterns according to the network energy, which means that the robot can generate gait patterns by connecting legs and transit gait patterns according to such parameters as the desired speed.

Keywords: Central pattern generator; Gait pattern; Pattern transition; Legged robot; Autonomous decentralized system

1. Introduction

Legged animals can generate walk patterns (gaits) suitable to their walking speed. The gait-generating system is achieved by the central pattern generator (CPG), a group of neurons located in a central nervous system (Fig. 1) [1]. The CPG is modeled as a system of coupled nonlinear oscillators because of its periodical output (oscillation patterns) and is used to coordinate leg movements (Fig. 1) [2–6].

The autonomous decentralized multi-legged robot system has a leg as an autonomous partial system (subsystem) [7,8]. Fig. 2 shows the autonomous decentralized multi-legged robot: NEXUS. Each subsystem has oscillators to control the leg movement. This robot system achieves the function of a CPG as a whole system by coupling with neighbor oscillators, i.e., by coupling with neighbor subsystems. Each subsystem does not need to discover the total number of subsystems or its position in the network. However, this robot system is expected to have properties such as a changeable number of legs, easy maintenance, failure tolerance, and environmental adaptability.

Many CPG models have been proposed in previous research. However, these models, which are divided

* Corresponding author. Tel.: +81-3-789-5335; fax: +81-3-789-2768.
E-mail addresses: inagaki@nuem.nagoya-u.ac.jp (S. Inagaki), arai@prince.pe.u-tokyo.ac.jp (T. Arai).
[1] Tel.: +81-3-5841-6457; fax: +81-3-5841-6487.

0921-8890/$ – see front matter © 2003 Elsevier Science B.V. All rights reserved.
doi:10.1016/S0921-8890(03)00067-8

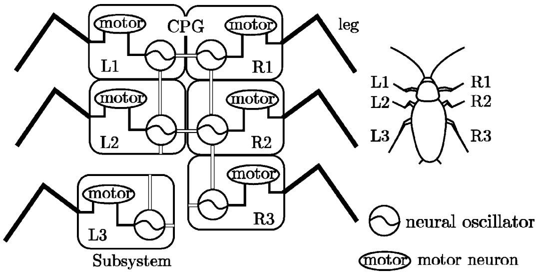


Fig. 1. Schema of a CPG: each leg is coordinated by the neural oscillator output via the motor neuron.

into two types [5], cannot deal with changes in the number of oscillators and the number of legs. One type requires a phase relation of common gait patterns according to the number of oscillators [2,3]. The other type is required to analyze how to change parameters and input in various cases [5,6].

The purpose of our research is the construction of a CPG model with the following features in an autonomous decentralized multi-legged robot system:

1. Gait patterns can be generated automatically according to the number of legs without embedded gait patterns.
2. Transitions in gait patterns can be generated by changing a certain parameter, e.g., the network energy.

This paper is about gait control while the number of legs (the configure of the robot) changes. There are pioneering researches regarding to the locomotion of the self-reconfiguration robot [9]. The method proposed in this paper is different from the previously proposed methods in two aspects. First, the locomotion is achieved from the relationship of the neighbor legs. The other, each leg does not need to know the topology (where the leg is in the robot). In this paper, a method for constructing a CPG model is proposed from the viewpoint of the graph expression of the oscillator network and the oscillator dynamics. The oscillator and the interaction correspond to a vertex and an edge, respectively (Fig. 3). Thereby, the oscillation patterns result in the eigen value problem of the graph and the transition is achieved only by changing the oscillation energy (network energy). Furthermore, the locality of the subsystem is realized due to the locality of the oscillator dynamics.

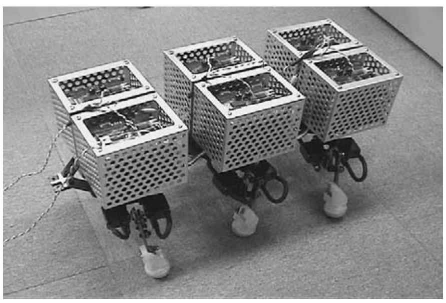


Fig. 2. Autonomous decentralized multi-legged robot: NEXUS.

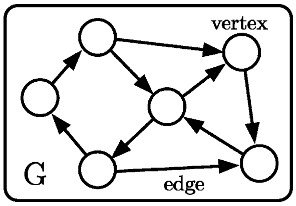


Fig. 3. Graph expression of a system.

The organization of this paper is as follows. In Section 2, a graph and a functional space are defined. The Hamilton system is introduced on the graph in Section 3, and the system is shown to obtain inherently multiple natural modes of oscillation. In Section 4, the method for constructing a CPG model is shown by using a simple example (hexapod). In Section 5, the Hamilton system is synthesized with a gradient system to select and transit the modes. Computer simulations of a hexapodal CPG model are demonstrated in Section 6.

2. Definition of a graph and function spaces

This section starts with some definitions of a complex function on a graph. Let a set of vertices be V and a set of edges be E. Then a graph G can be defined as a set of these two terms $G = (V, E)$ (Fig. 3).

Generally, vertices in the graph correspond to the agents of an autonomous decentralized system. In the case of the CPG model, they correspond to the oscillators.

It is assumed that G is a finite and oriented graph with N vertices. That is, the number of edges is finite.

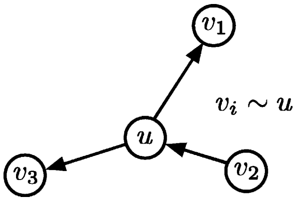


Fig. 4. Vertex u and its neighbor vertices v.

Each edge connects the initial vertex to the terminal one because the co-differentiation or gradient is defined on the graph [10,11]. Note that this direction is not related with to information flow or the interaction of the vertices.

Each vertex $u \in \mathrm{V}$ has a complex variable $\psi(u) = q(u) + \mathrm{i}p(u)$ for $q(u),\ p(u) \in \mathbf{R}$. Let $C(\mathrm{V})$ be the set of all complex (or real) functions on V. Then $\boldsymbol{\psi} = (\psi(u)_{u\in \mathrm{V}}) \in C(\mathrm{V})$ is defined as the group of $\psi(u)\,\forall\, u \in \mathrm{V}$. $\boldsymbol{\psi}$ is also defined as $\boldsymbol{\psi} = \mathbf{q} + \mathrm{i}\mathbf{p}$ for $[\mathbf{q}, \mathbf{p}]^{\mathrm{T}} = [(q(u)_{u\in\mathrm{V}}), (p(u)_{u\in\mathrm{V}})]^{\mathrm{T}}$.

The Laplacian operator Δ_A on a finite graph G is defined for $\boldsymbol{\psi} \in C(\mathrm{V})$ as [10]

$$(\Delta_A \boldsymbol{\psi})(u) = \sum_{v\sim u} (\psi(u) - \psi(v)). \tag{1}$$

$v \sim u$ shows the set of vertices in which vertex v connects to vertex u (Fig. 4). The value of Eq. (1) means Laplacian value. Laplacian value shows the difference between the vertex value $\psi(u)$ and the average of neighbor vertex values $\psi(v)$. Moreover, Laplacian value is independent of the direction of edges. Note that Δ_A is calculated as $\Delta_A = A_a A_a^{\mathrm{T}}$, where A_a is an incidence matrix of G [12]. This incidence matrix A_a is determined by connections of vertices.

3. The Hamilton system on a graph

3.1. Wave equation as the Hamilton system

A wave equation is introduced on the graph G, which is part of the Hamilton system. When a wave equation is used, the discrete natural frequency modes correspond to the gait patterns. Moreover, the discrete energy states correspond to the discontinuous walk speeds.

Schrödinger's wave equation is introduced as a concrete wave equation. The reason is as follows:

1. A complex variable is useful when we express the vibrating state.
2. We can use the derivation in quantum mechanics as for the Hamiltonian value and their natural frequency modes.
3. There is a symmetry between the real part and the imaginary part in the dynamics. The symmetry makes each variable's trajectory a circle in the complex plane.

Let $\boldsymbol{\psi}(t) = (\psi(u,t))_{u\in\mathrm{V}} \in C(\mathrm{V}\times\mathrm{R})$ be a wave function on graph G. Then Schrödinger's wave equation in graph G is defined as

$$\mathrm{i}h \frac{\partial \boldsymbol{\psi}(t)}{\partial t} = \mathcal{H}\boldsymbol{\psi}(t), \tag{2}$$

where h plays the role of Planck's constant, which connects the Hamiltonian value H and the angular velocity. Hamiltonian operator $\mathcal{H}$ is defined as $\mathcal{H} \equiv \Delta_A$. Then each dynamic of Eq. (2) at each vertex is expressed as

$$\frac{\partial q(u,t)}{\partial t} = \frac{1}{h}\sum_{v\sim u}(p(u,t) - p(v,t)), \tag{3}$$

$$\frac{\partial p(u,t)}{\partial t} = -\frac{1}{h}\sum_{v\sim u}(q(u,t) - q(v,t)). \tag{4}$$

This equation shows that the dynamic at each vertex is determined by variables of neighbor vertices $v \sim u$ and itself u (Fig. 4).

The total Hamiltonian value H of this Hamiltonian operator $\mathcal{H}$, shown in Eq. (5), is defined as the energy of this network (H is called the total Hamiltonian value). The total Hamiltonian value H can be decomposed as the local value $H(u)$ at each vertex u:

$$H = \boldsymbol{\psi}(t) \cdot \mathcal{H}\boldsymbol{\psi}(t) = \sum_{u\in\mathrm{V}} H(u), \tag{5}$$

$$H(u) = \sum_{v\sim u}(q^2(u,t) + p^2(u,t) - q(u,t)q(v,t) - p(u,t)p(v,t), \tag{6}$$

where $\boldsymbol{\psi}_1(t) \cdot \boldsymbol{\psi}_2(t)$ is an inner product of $\boldsymbol{\psi}_1(t)$ and $\boldsymbol{\psi}_2(t)$. $H(u)$ is also determined by variables of neighbor vertices and itself, so $H(u)$ is called the local Hamiltonian value of u.

3.2. Natural frequency modes of wave

Natural frequency modes of the wave equation (2) are solutions of the following equation:

$$\mathcal{H}\boldsymbol{\psi}(t) = H\boldsymbol{\psi}(t). \tag{7}$$

That is, the natural frequency modes $\boldsymbol{\psi}_m(t)$ $(m = 0, 1, 2, \dots, N-1)$ are provided as eigen functions of the Hamiltonian operator $\mathcal{H}$. Then the total Hamiltonian value H_m is given as the eigen value of the Hamiltonian operator $\mathcal{H}$.

With the separation of variables, these natural frequency modes of Eq. (7) can be expressed by the spatio-function $\boldsymbol{\varphi}_m = (\varphi_m(u))_{u\in V} \in C(\mathrm{V})$, where $\|\boldsymbol{\varphi}_m\| = 1$ and the temporal-function $\phi_m(t) = \mathrm{e}^{-\mathrm{i}\omega_m t} \in \mathbf{R}$:

$$\boldsymbol{\psi}_m(t) = \boldsymbol{\varphi}_m\phi_m(t) = \boldsymbol{\varphi}_m\,\mathrm{e}^{-\mathrm{i}\omega_m t}, \tag{8}$$

where ω_m is the natural angular velocity, which is related to the total Hamiltonian value H_m as $H_m = h\omega_m$. In this Hamilton system, an arbitrary wave function $\psi(t)$ is expressed by the linear sum of the natural frequency mode:

$$\boldsymbol{\psi}(t) = \sum_{m=0}^{N-1} k_m\boldsymbol{\psi}_m(t), \tag{9}$$

where $k_m \in \mathbf{C}$ are the mode coefficients.

4. Example: graph with six vertices

4.1. Natural oscillation modes that appear in a graph of six vertices

In this section, we consider a graph composed of six vertices as shown in Fig. 5. In this case, the incidence matrix A_a is as follows:

$$A_a = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}. \tag{10}$$

In the incidence matrix, the row corresponds to the vertex and the line corresponds to the edge. ‘1’ is allocated when the directed edge comes out of the vertex, and conversely ‘−1’ is allocated when getting in.

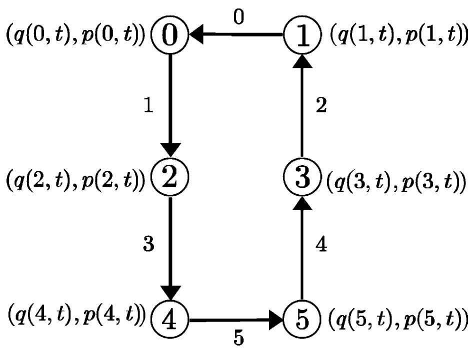


Fig. 5. Graph of six vertices.

And then, the Hamiltonian operator of Eq. (2) becomes

$$\mathcal{H} = \Delta_A = \begin{bmatrix} 2 & -1 & -1 & 0 & 0 & 0 \\ -1 & 2 & 0 & -1 & 0 & 0 \\ -1 & 0 & 2 & 0 & -1 & 0 \\ 0 & -1 & 0 & 2 & 0 & -1 \\ 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & -1 & -1 & 2 \end{bmatrix}. \tag{11}$$

The eigenfunctions

$$\boldsymbol{\varphi}_m = [\varphi_m(0), \varphi_m(1), \dots, \varphi_m(5)]^{\mathrm{T}}$$

and total Hamiltonian values H_m are obtained from Eq. (7) as shown in Table 1.

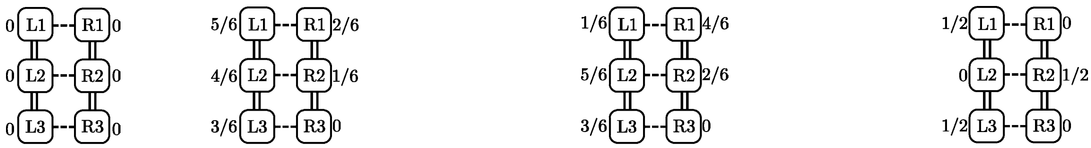
4.2. Construction of the CPG model

Here, an N-legged robot (N is an even number) has a CPG model composed of $2N$ oscillators as shown in Fig. 6 [8,13]. Two networks are symmetrically arranged, and oscillators $0, 2, \dots, N-2$ and $0', 2', \dots, (N-2)'$ control timings of the legs L1, L2, …, L$(N/2)$ and R1, R2, …, R$(N/2)$, respectively.

Here, it is assumed that the phases of the wave on one side are a half-period earlier (or later) than those of the wave on the other side. Then common hexapodal gait patterns of insects can be applied (Table 1) [14]. The reasons for employing the structure are (1) the

Table 1
The solution of the eigenvalue problem and hexapodal gaits (phase relations) according to each mode (one cycle is set to be 1 on the basis of the right rear leg)

m	0	1	2	3	4	5
H_m	0	1	1	3	3	4
$\boldsymbol{\varphi}_m$	$\frac{\sqrt{6}}{6}\begin{bmatrix}1\\1\\1\\1\\1\\1\end{bmatrix}$	$\frac{1}{2}\begin{bmatrix}-1\\0\\-1\\1\\0\\1\end{bmatrix}$	$\frac{1}{2}\begin{bmatrix}0\\-1\\1\\-1\\1\\0\end{bmatrix}$	$\frac{1}{2}\begin{bmatrix}1\\0\\-1\\-1\\0\\1\end{bmatrix}$	$\frac{1}{2}\begin{bmatrix}0\\1\\-1\\-1\\1\\0\end{bmatrix}$	$\frac{\sqrt{6}}{6}\begin{bmatrix}-1\\1\\1\\-1\\-1\\1\end{bmatrix}$
Gait	(a) stand	(b) metachronal tripod		(c) rolling tripod		(d) tripod



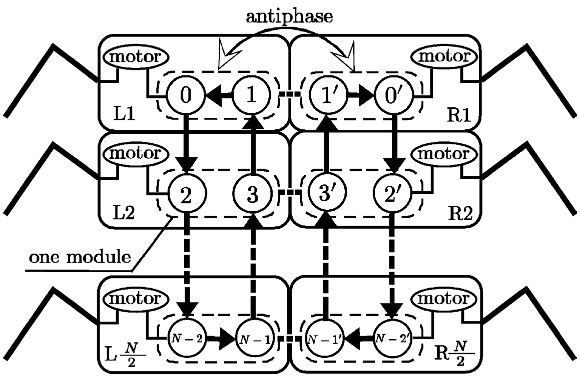


Fig. 6. The structure of an N-legged robot: a region in a broken oval is treated as one module.

generation of waves traveling in the same direction on both sides and (2) the locality of the connections between oscillators. For example, the CPG model can generate a 'metachronal tripod' in hexapodal gaits and 'walk'[2] in quadrupedal gaits.

The natural oscillation modes obtained from Fig. 5 are matched with hexapodal gaits as shown in Table 1. The natural modes with the same eigen value form one gait by exchanging themselves through one oscillation cycle. This oscillation pattern is called an exchange pattern, e.g., 'exchange pattern 1&2' or 'exchange pattern 3&4.' Interestingly, the larger the total Hamiltonian value H_m becomes, the more rapid these gaits are in real insects.

[2] The step order is left-forward, right-hind, right-forward, left-hind.

5. Synthesis of the Hamilton system and a gradient system

The Hamilton system merely produces a linear sum of the oscillating modes as we demonstrated in Section 3. The dynamic (a) and (b) is realized via the synthesis of the Hamilton system and a gradient system:

(a) One (or two) specific natural oscillation mode (or modes) appears alternatively out of multiple natural oscillation modes.
(b) Changes between gait patterns are performed properly by changing a certain parameter, such as the walk speed.

The procedures necessary to realize (a) and (b) are considered as follows (Fig. 7):

(a) The variable $\psi(u, t)$ of each vertex converges to a circular orbital unit in a complex plane. Then the natural oscillation mode appears, whose set of the

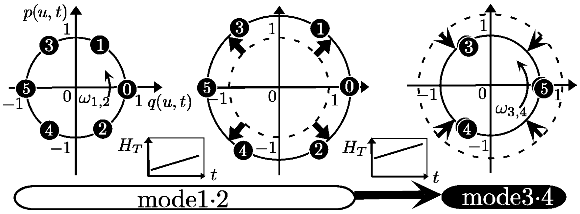


Fig. 7. Transition between natural oscillation modes with the target value H_{T}.

local Hamiltonian value is the closest to the local Hamiltonian target value H_{T}.

(b) When the target value H_{T} of local Hamiltonian value $H(u)$ increases (or decreases) continuously, the variable of each vertex follows it by enlarging (or lessening) the orbital radius at first. When the amount of the radius change becomes even larger, the variables of the vertices change (bifurcate) the natural oscillation mode and continue the follow-up to the target value H_{T}.

Let

$$[\mathbf{x}_1(t), \mathbf{x}_2(t)]^{\mathrm{T}} = [(x_1(u,t))_{u\in \mathrm{V}}, (x_2(u,t))_{u\in \mathrm{V}}]^{\mathrm{T}}$$

be the function in graph G. If a potential function is represented as $P(\mathbf{x}_1(t), \mathbf{x}_2(t))$, the dynamics of the gradient system [11] can be expressed at each vertex $u \in \mathrm{V}$ as

$$\frac{\partial x_1(u,t)}{\partial t} = -\frac{\partial P(\mathbf{x}_1(t), \mathbf{x}_2(t))}{\partial x_1(u,t)}, \tag{12}$$

$$\frac{\partial x_2(u,t)}{\partial t} = -\frac{\partial P(\mathbf{x}_1(t), \mathbf{x}_2(t))}{\partial x_2(u,t)}. \tag{13}$$

This dynamics is effectual in converging $[\mathbf{x}_1(t), \mathbf{x}_2(t)]^{\mathrm{T}}$ to the state in which $P(\mathbf{x}_1(t), \mathbf{x}_2(t))$ can be minimized. The potential function $P(\mathbf{x}_1(t), \mathbf{x}_2(t))$ is given as the sum of local potential functions as shown in

$$P(\mathbf{x}_1(t), \mathbf{x}_2(t)) = \sum_{u\in \mathrm{V}} P(u), \tag{14}$$

where $P(u)$ is determined by the variables of neighbor vertices $v \sim u$ and itself u. From Eqs. (3) and (4)) and Eqs. (12) and (13), each vertex dynamic of the synthesized system can be expressed as

$$\frac{\partial q(u,t)}{\partial t} = \frac{1}{h}\sum_{v\sim u}(p(u,t) - p(v,t)) - \frac{\partial P(\mathbf{q}(t), \mathbf{p}(t))}{\partial q(u,t)}, \tag{15}$$

$$\frac{\partial p(u,t)}{\partial t} = -\frac{1}{h}\sum_{v\sim u}(q(u,t) - q(v,t)) - \frac{\partial P(\mathbf{q}(t), \mathbf{p}(t))}{\partial p(u,t)}. \tag{16}$$

In this research, the potential function is chosen from (a) and (b) as

$$P(u) = \alpha P_0(u) + \beta P_1(u) + \gamma P_2(u), \tag{17}$$

$$P_0(u) = q^2(u,t) + p^2(u,t) + \frac{1}{q^2(u,t) + p^2(u,t)}, \tag{18}$$

$$P_1(u) = (H(u) - H_{\mathrm{T}})^2, \tag{19}$$

$$P_2(u) = P_2(u') = (q(u,t) + q(u',t))^2 + (p(u,t) + p(u',t))^2, \tag{20}$$

where α, β and γ are real constants, and H_{T} the target value of the local Hamiltonian value $H(u)$. $P_0(u)$ works so that the variable of each vertex converges to a circular orbital with a radius 1 in the complex plane. $P_1(u)$ works so that the local Hamiltonian value $H(u)$ of each vertex follows the target value H_{T}. $P_2(u)$ works so that waves in the left and right networks become antiphased.

The method of constructing the gradient system in the graph is the same as the one used by Yuasa and Ito [11]. The significant difference is that the method of Yuasa and Ito [11] requires embedding oscillation patterns beforehand (as phase differences of the oscillators) and the proposed method obtains oscillation patterns which are generated from the connection of the oscillator network due to the introduction of the Hamilton system.

Each dynamic of Eqs. (15) and (16) at each vertex is expressed as follows:[3]

$$\begin{aligned}\frac{\partial q(u)}{\partial t} = {} & \frac{1}{h}\sum_{v\sim u}(p(u) - p(v)) \\ & - 2a\left\{q(u) - \frac{q(u)}{(q^2(u) + p^2(u))^2}\right\} \\ & - 2b\sum_{v\sim u}\{(H(u) - H_{\mathrm{T}})(2q(u) - q(v)) \\ & \qquad + (H(v) - H_{\mathrm{T}})(-q(v))\} \\ & - 4c(q(u) + q(u')),\end{aligned} \tag{21}$$

[3] t is omitted.

$$\frac{\partial p(u)}{\partial t} = -\frac{1}{h}\sum_{v\sim u}(q(u)-q(v)) - 2a\left\{p(u) - \frac{p(u)}{(q^2(u)+p^2(u))^2}\right\} - 2b\sum_{v\sim u}\{(H(u)-H_{\mathrm{T}})(2p(u)-p(v)) + (H(v)-H_{\mathrm{T}})(-p(v))\} - 4c(p(u)+p(u')). \quad (22)$$

These equations show that the dynamic at each vertex is determined by values of neighbor vertices $q(v,t)$, $p(v,t)$, $H(v,t)$, $q(u',t)$ and $p(u',t)$. The purpose of this paper is to achieve a globally harmonious motion from the relationship of the neighbor oscillators. With the simulation of the next paragraph, it is proved that these dynamics (21) and (22) can create the globally harmonious motion: the gait-pattern generation and transition.

On the contrary, just the target value H_{T} is a common value in all oscillators. It is sent to all oscillators from the upper computing system or the operator by the broadcast message. The walking speed is decided by the target value H_{T}. Of course, it depends on how to create the leg motion from the variables of the oscillator. For example, the oscillator norm and angular are converted to the step length and phase, respectively.

6. Gaits generation and transition in computer simulations

We simulated generation and transition of hexapodal gaits with the connection as shown in Fig. 6. The parameters in Eq. (17) were $\alpha = 1.0$, $\beta = 0.1$, $\gamma = 0.05$, and the initial variables $\mathbf{q}(0)$, $\mathbf{p}(0)$ were generated, respectively, by a uniform random number of $[-1, 1]$.

The target value of the local Hamiltonian H_{T} was

$$H_{\mathrm{T}} = 6\left(1 - \cos\left(\frac{2\pi t}{400}\right)\right) \quad (23)$$

as shown in Fig. 8 (broken line). The results of this simulation ($t = 0$–400) are shown in Figs. 8 and 9.

Fig. 8 shows the changes of the total Hamiltonian value H (solid line) and the target value of the local Hamiltonian H_{T} (broken line). H_{T} is given to all vertices as the convergence target value of the local Hamiltonian value $H(u)$. H is calculated by Eq. (5).

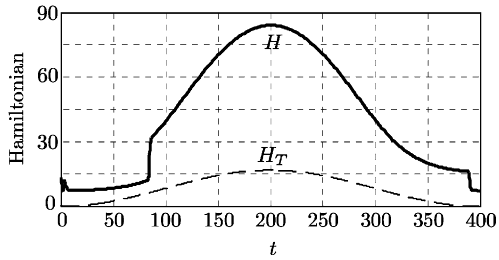


Fig. 8. Total Hamiltonian H of the left network of Fig. 6 and the target value of local Hamiltonian H_{T}.

H dose not absolutely follow H_{T} because the system follows the change of H_{T} by changing the oscillation modes (Section 5, (a) and (b)). The oscillation pattern transited from the exchange pattern 1&2 to 3&4 at around $t = 80$ and from the exchange pattern 3&4 to 1&2 at around $t = 390$. As shown in Fig. 8, H changed dramatically between the exchange patterns 1&2 and 3&4.

Fig. 9 shows a plot of real parts of $\psi(u,t)$ ($u = 0$, 2, 4 and $0'$, $2'$, $4'$) at $t = 380$–400. The oscillation pattern transited from the exchange pattern 3&4 to 1&2 at around $t = 390$. The number above each curve expresses the number of the corresponding oscillator. The former and latter oscillation patterns correspond to the rolling tripod and the metachronal tripod, respectively, from Table 1.

The proposed CPG model can achieve transitions between oscillating patterns against the change of H_{T}. Furthermore, if the target value of the local Hamiltonian H_{T} is matched with the desired walk speed, the gait patterns can be changed accordingly.

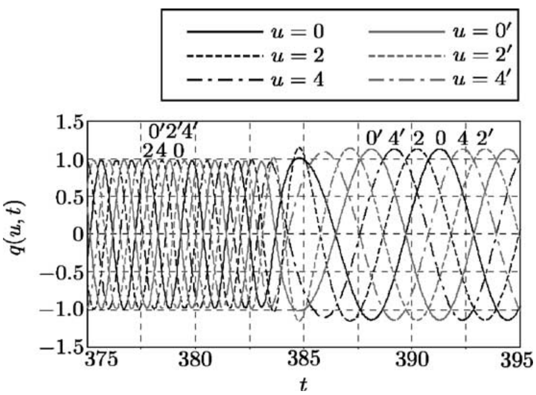


Fig. 9. The wave forms of $p(u,t)(u = 0, 2, 4, 0', 2', 4')$.

When the number of legs is changed, the closed loops are formed in the left and right oscillator networks by connecting two oscillators in the foremost legs and the last legs. The extraction and transition of modes are also observed in the cases of 4-legged and 8-legged CPG models with the same parameters. However, they depend on the initial condition to some degree because the potential functions of Eqs. (17)–(20) were chosen with the heuristics method. We are now establishing a methodology for designing a suitable potential function.

7. Discussion

In this paper, a method for constructing a mathematical CPG model was proposed. The dynamic of each oscillator is determined only by information about itself and neighbor oscillators. Nevertheless, the total system has natural oscillation modes as the total-order (gait pattern). The order depends on a scale of the total system, i.e., the number of oscillators. As a result, we can produce a CPG model adequate for an actual robot NEXUS (Section 1, points 1. and 2.).

The influence on the CPG model of the body characteristics is important when the robot actually walks [7,15–17]. In regard to this, feedback to the CPG model from the robot's sensors is required. However, this paper does not deal with feedback. In our method, an open-loop system is built first:

Input target speed (desired value of the local Hamiltonian value);
State wave function;
Output oscillating pattern.

We introduce the feedback to the CPG model from sensors. Specifically, we consider

- the correspondence of the actual walk speed and the target value of local Hamiltonian H_{T};
- the change of the Hamiltonian operator $\mathcal{H}$ of Eq. (2) to the sensor information;
- the addition of new potential functions to Eq. (17).

Moreover, we consider how each leg moves against the changing variables of the oscillator. The correspondence is required to carry the CPG model in the actual robot. We are now installing the CPG model to the actual multi-legged robot (Fig. 2).

References

[1] K. Pearson, The control of walking, Scientific American 235 (6) (1976) 72–86.
[2] G. Schöner, W.Y. Jiang, J.A.S. Kelso, A synergetic theory of quadruped gaits and gait transitions, Journal of Theoretical Biology 142 (1990) 359–391.
[3] H. Yuasa, M. Ito, Coordination of many oscillators and generation of locomotory patterns, Biological Cybernetics 63 (1990) 177–184.
[4] J.S. Bay, H. Hemami, Modeling of a neural pattern generator with coupled nonlinear oscillators, IEEE Transactions on Biomedical Engineering BME-34-4 (1987) 297–306.
[5] J.J. Collins, S.A. Richmond, Hard-wired central pattern generators for quadrupedal locomotion, Biological Cybernetics 71 (1994) 375–385.
[6] C.C. Canavier, R.J. Butera, R.O. Dror, D.A. Baxter, J.W. Clark, J.H. Byrne, Phase response characteristics of model neurons determine which patterns are expressed in a ring circuit model of gait generation, Biological Cybernetics 77 (1997) 367–380.
[7] K. Tujita, K. Tuchiya, A. Onat, S. Aoi, M. Kawakami, Locomotion control of a multipod locomotion robot with CPG principles, in: Proceedings of the Sixth International Symposium of Artificial Life and Robotics, vol. 2, 2001, pp. 421–426.
[8] T. Odashima, H. Yuasa, Z.W. Luo, M. Ito, Emergent generation of gait pattern for a myriapod robot system based on energy consumption, Journal of the Robotics Society of Japan 17 (8) (1999) 1149–1157 (in Japanese).
[9] W.-M. Shen, B. Salemi, P. Will, Hormone-inspired adaptive communication and distributed control for CONRO self-reconfigurable robots, IEEE Transactions on Robotics and Automation 18 (5) (2002).
[10] H. Urakawa, Laplacian and Networks, Shokabo, Tokyo, 1996 (in Japanese).
[11] H. Yuasa, M. Ito, Autonomous decentralized systems and reaction–diffusion equation on a graph, Transactions of the Society of Instrument and Control Engineers 35 (1999) 1447–1453 (in Japanese).
[12] T. Ozawa, Electric Circuit I, Shokodo, Tokyo, 1980 (in Japanese).
[13] M. Golubitsky, I. Stewart, P.L. Buono, J.J. Collins, A modular network for legged locomotion, Physica D 115 (1998) 56–72.
[14] D.M. Wilson, Insect walking, Annual Review of Entomology 11 (1966) 103–122.
[15] M. Yano, M. Tokiwa, Y. Makino, Simultaneous self-organization of optimal gait pattern of insect and trajectory to its destination, in: Proceedings of the Fourth IFAC Symposium on Intelligent Autonomous Vehicles, 2001, pp. 261–264.
[16] S. Ito, H. Yuasa, M. Ito, An adaptation theory in autonomous decentralized systems, Transactions of the Society of Instrument and Control Engineers 35 (5) (1999) 684–692 (in Japanese).
[17] H. Kimura, S. Akiyama, K. Sakurama, Realization of dynamic walking and running of the quadruped using neural oscillator, Autonomous Robots 7 (3) (1999) 247–258.

Shinkichi Inagaki received the B.S. and M.S. degrees in Engineering from Nagoya University, Nagoya, Japan, in 1998 and 2000, respectively, and Ph.D. degree in Engineering from The University of Tokyo, Tokyo, Japan, in 2003. Since 2003, he has been the Research Associate at the Department of Electronic-Mechanical Engineering, Nagoya University. His specialties are autonomous decentralized systems and the application to robotics. He is a Member of SICE, JSME and JRS.

Hideo Yuasa received the B.S. and M.S. degrees in Engineering from Nagoya University in 1984 and 1986, respectively. He obtained Ph.D. from the university in 1992 and worked for the Department of Electronic-Mechanical Engineering, Nagoya University and Bio-Mimetic Control Research Center, RIKEN. Since April 1999, he has been the Associate Professor at the Department of Precision Engineering, The University of Tokyo.

His specialty was distributed autonomous systems, especially mathematical models and its application. He was famous in fundamental models of autonomous systems. He was a Program Co-chair of IAS-7. He died suddenly in September 2000.

Tamio Arai graduated from The University of Tokyo in 1970 and obtained Dr. of Engineering from the same university. He has been a Professor in Department of Precision Engineering since 1987 and now the Director of Research into Artifacts, Center for Engineering as a Director, in The University of Tokyo. His specialties are assembly and robotics, especially multiple mobile robots including legged robot league of RoboCup. He works in IMS programs in Holonic Manufacturing System. He is a Member of CIRP, IEEE/RAS, RSJ, JSPE, and Honorary President of JAAA.